SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304
Course Title: AI Assisted Coding
Year/Sem: III / II
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Lab Objectives

- Write advanced SQL queries using AI assistance
- Understand joins, aggregation, and subqueries
- Improve query performance
- Analyze real-world database problems

Task 1: JOIN Operations

Tables

```
Students(id, name)
Courses(id, course_name)
Enrollments(student_id, course_id)
```

Query

```
SELECT Students.name, Courses.course_name
FROM Students
JOIN Enrollments ON Students.id = Enrollments.student_id
JOIN Courses ON Courses.id = Enrollments.course_id;
```

Task 2: Aggregate Functions

```
SELECT course_id, COUNT(*) AS total_students
FROM Enrollments
GROUP BY course_id;
```

Task 3: Subqueries

```
SELECT name
FROM Students
WHERE id IN (
    SELECT student_id FROM Enrollments
);
```

Task 4: Nested Queries

```
SELECT name
FROM Students
WHERE id = (
    SELECT student_id
    FROM Results
    ORDER BY marks DESC
    LIMIT 1
);
```

Task 5: Performance Optimization

```
CREATE INDEX idx_student_id ON Enrollments(student_id);

EXPLAIN SELECT * FROM Enrollments WHERE student_id = 1;
```

Conclusion

Advanced SQL techniques improve data retrieval efficiency.

AI tools help generate optimized queries, but understanding query logic is essential.